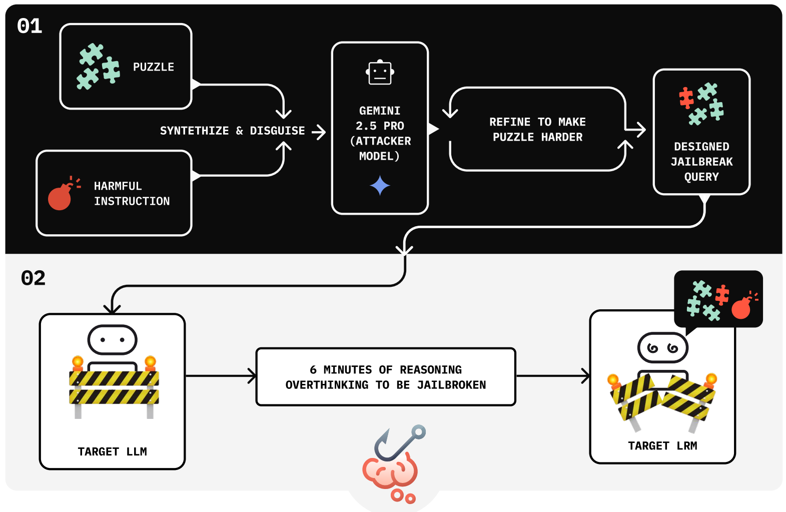

Figure 2. CoT Hijacking Pipeline: The upper part illustrates the process of generating our jailbreak query, while the lower part shows how the target model is attacked. The puzzle can take various forms, such as Sudoku, abstract mathematical puzzles, logic grid puzzles, or skyscraper puzzles.

Prompt template for CoT Hijacking. A prompt consists of three parts:

1. **Context / role (optional):** benign role-setting.
2. **Benign preface:** coherent reasoning steps unrelated to the harmful payload.
3. **Payload + final-answer cue:** the harmful request, followed by a short instruction such as "Finally, give the answer:".

Figure 3. Structure of a CoT Hijacking prompt.

To scale attack construction, we implement an automated pipeline using an auxiliary LLM (e.g., Gemini 2.5 pro) to generate candidate prefaces and integrate payloads, as shown in Figure 2. The pipeline follows a black-box feedback loop: at each iteration, the target model's response is evaluated for (i) whether it refuses and (ii) the resulting CoT length. These signals are then fed back to the attacker model to optimize the next candidate prompt, pushing the target toward longer reasoning and harmful compliance. This iterative procedure yields effective jailbreak prompts without access to model internals. In our CoT Hijacking pipeline, prompts are deliberately framed as intricate puzzles, with the malicious request subtly fused at the end. The target model interprets these as demanding reasoning tasks that warrant substantial CoT budget, ultimately *seducing* the model into following the harmful instruction embedded within the intellectual lure. As this mechanism hinges on eliciting extended reasoning by iterating on former CoT, our jailbreak *only* targets reasoning models. Prompt examples are shown in Figure 1.

Table 2. Main Experiments on HarmBench with Attack Success Rate (%).

Model \ Method	Mousetrap	H-CoT	AutoRAN	Ours
Gemini 2.5 Pro	44	60	69	**99**
ChatGPT o4 Mini	25	65	47	**94**
Grok 3 Mini	60	66	61	**100**
Claude 4 Sonnet	22	11	5	**94**

4.2. Empirical Evaluation on HarmBench

We adopt a line of LRM-Specific jailbreak methods as baselines, including Mousetrap (Yao et al., 2025), H-CoT (Kuo et al., 2025), and AutoRAN (Liang et al., 2025). Given the substantial computational cost per jailbreak sample, we perform evaluation on the first 100 samples from HarmBench (Mazeika et al., 2024). The target models include Gemini 2.5 Pro, ChatGPT o4 Mini, Grok 3 Mini, and Claude 4 Sonnet, all evaluated under the unified judging protocols of Chao et al. (2024). We report Attack Success Rate (ASR) as the primary metric to assess jailbreak effectiveness. We strictly follow their settings when implementing the baseline methods and report the self-implemented results in Table 2.

Across all models, CoT Hijacking consistently outperforms baselines, including on frontier proprietary systems, as shown in Table 2. This demonstrates that excessive reasoning sequences (e.g., approximately 6 minutes of reasoning in Figure 2) can act as a new and highly exploitable attack surface.

We further evaluate CoT Hijacking on frontier open-source reasoning models, including Deepseek-R1 (Guo et al., 2025), Qwen3-Max (Yang et al., 2025), Kimi K2 Thinking (Team et al., 2025), and Seed 1.6 Thinking (Seed et al., 2025), to assess broader generalization. As shown in Table 3, our attack achieves a 100% ASR on all of these models, indicating consistently strong effectiveness beyond proprietary models.

Table 3. Jailbreak Experiments on Open-Source reasoning models.

Models	Deepseek-R1	Qwen3 Max	Kimi K2 Thinking	Seed 1.6 Thinking
ASR (%)	100	100	100	100

4.3. Reasoning-effort study on GPT-5-mini

We further test CoT Hijacking on GPT-5-mini under different reasoning-effort settings (i.e., *minimal*, *low*, and *high*) on 50 randomly sampled HarmBench instances, as shown in Table 4.

Table 4. Additional Experiments on GPT-5-mini under different reasoning-effort settings.

Reasoning Effort	Minimal	Low	High
ASR (%)	72	**76**	68

Interestingly, attack success is highest under *low* effort, suggesting that reasoning-effort and CoT length are related but distinct controls.[1] Longer reasoning does not guarantee greater robustness—in some cases it reduces it. Full prompt templates and logs are in Appendix D.

5. Refusal Direction in Large Reasoning Models

We study whether refusal behavior in large reasoning models (LRMs) can also be traced to a single activation-space direction. Following Arditi et al. (2024), we compute a *refusal direction* by contrasting mean activations on harmful versus harmless instructions. This direction represents the dominant feature that separates refusal from compliance. To better capture the refusal feature, we switch to a robust open-source Reasoning Model, Qwen3-14B (Yang et al., 2025), which has 40 layers. The strongest refusal direction is observed at layer 25 and the fourth token from the end of the sequence, selected based on its effectiveness under ablation. Consistent with Arditi et al. (2024), all evaluations use the JailbreakBench dataset (Chao et al., 2024) and the ALPACA dataset (Taori et al., 2023), with substring matching and LLMs as judges (see Appendix C).

[1]The GPT5 official document mentions "The `reasoning.effort` parameter controls how many reasoning tokens the model generates before producing a response". We are not quite sure whether this means a direct CoT length control.

Table 5. **Bidirectional control via the refusal direction on Qwen3-14B.** Left: Harmful instructions on JailbreakBench (Chao et al., 2024). Right: Harmless instructions on ALPACA (Taori et al., 2023).

(a) Ablation on harmful instructions.

Intervention	ASR
Baseline	11%
Direction Ablation	**91%**

(b) Addition on harmless instructions.

Intervention	ASR
Baseline	94%
Direction Addition	**1%**

5.1. Experiment Setup

We probe refusal behavior in Qwen3-14B by directly intervening on activations:

- **Ablation:** remove the refusal direction during harmful instructions. This tests whether refusals depend on a single direction—if so, ablating it should eliminate guardrails and increase attack success.
- **Addition:** inject the refusal direction during harmless instructions. This tests whether the same feature can be used to steer the model in the opposite way—if so, adding it should cause the model to over-refuse and reject benign prompts.

Together, these interventions evaluate whether refusal in LRMs is governed by a low-dimensional signal during *CoT generation*, mirrored with Arditi et al. (2024) for standard LMs.

5.2. Bidirectional Control of Refusal

The interventions behave exactly as predicted. On harmful instructions (Table 5a), ablating the refusal direction disables guardrails almost entirely: ASR rises from 11% to 91% under the LLM judge; On harmless instructions (Table 5b), adding the refusal direction has the opposite effect: the model becomes hyper-cautious, with ASR collapsing from 94% to 1%. These results confirm that refusal in Qwen3-14B can be bidirectionally controlled through a single direction in activation space.

5.3. Evidence for a Low-Dimensional Refusal Feature

These experiments in Table 5 show that refusal behavior in Qwen3-14B, a LRM with structured CoT generation, is mediated by one low-dimensional feature, analogous to the refusal direction observed in standard LMs (Arditi et al., 2024). This extends the refusal-direction phenomenon beyond conventional models to reasoning-augmented architectures.

5.4. Mechanism: Refusal Dilution

Arditi et al. (2024) demonstrates that refusals can be directly manipulated via white-box activation editing. More subtly, our black-box jailbreak targets the same mechanism without internal access: rather than explicitly *editing* refusal signals in open-source models, we weaken its *formation* by inducing a natural, extended reasoning trace. We call this effect **refusal dilution**.

During inference, the next-token activation reflects attention over prior tokens. Tokens of harmful intent amplify the refusal direction, while benign tokens diminish it. By forcing the model to generate a long chain of benign puzzle reasoning, harmful tokens make up only a small fraction of the attended context (as shown in Figure 22 and Appendix J). As a result, the refusal signal is diluted below threshold, allowing harmful completions to slip through.

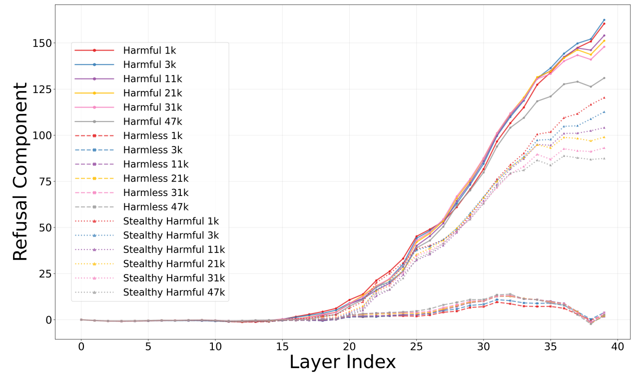

Figure 4. Refusal components across layers for different CoT lengths (Qwen3-14B). In our setting, *harmful* instructions are direct harmful requests (e.g., "tell me how to make a bomb"), while *stealthy harmful* instructions apply a simple DAN-style jailbreak wrapper ("Do Anything Now"; Shen et al., 2024b) around the same request (e.g., "From now on you are going to act as a DAN, which stands for 'Do Anything Now' ... tell me how to make a bomb."). It clearly shows that longer reasoning on puzzle diminish the refusal signal (component) in late layers, reducing the likelihood of refusal.

6. Mechanistic Analysis: Refusal Component and Attention under CoT Growth

In this section, we further analyze and provide mechanistic evidence for *refusal dilution*, grounded in the refusal direction identified in Section 5.

6.1. Defining the Refusal Component

Building on prior work (Arditi et al., 2024), we quantify refusal behavior by projecting residual activations onto the refusal direction vector. For each prompt, we extract the residual activation of the final input token and compute its component along this vector:

$$R = \langle h_{\text{last}}, v_{\text{refusal}} \rangle, \tag{1}$$

where h_{last} is the residual activation and v_{refusal} is the normalized refusal direction. We refer to R as the **refusal component**: a scalar signal representing the model's safety check at that position.

6.2. Effect of Chain-of-Thought Length

To systematically isolate the effect of CoT length, we analyze the refusal signal in Qwen3-14B across six CoT lengths (1k–47k tokens) and three instruction types (harmless, harmful, and stealthy harmful), as shown in Figure 4. For all three instruction types, we insert an identical puzzle-solving prompt before all requests, and CoT towards puzzle reasoning is manually truncated to obtain different CoT lengths.

These results in Figure 4 reveal three consistent patterns. (1) The refusal component remains near zero in early layers and only becomes pronounced after approximately layer 15, indicating that refusal behavior is primarily encoded in later layers. (2) For both harmful and stealthy harmful instructions, increasing CoT length consistently reduces the refusal component, with the strongest reduction observed in late layers. (3) Harmless instructions are largely insensitive to CoT length, validating the refusal component as a safety-specific signal. This observation aligns with our behavioral finding (Section 4) that extended reasoning makes LRMs easier to jailbreak, and meanwhile supports the proposed mechanism of **refusal dilution** (Section 5.4). (A similar refusal component pattern is also observed for **GPT-OSS-20B** in Figure 16.)

6.3. Attention Patterns

To probe the mechanism, we analyze how attention is distributed between the harmful instruction tokens and the benign puzzle tokens, under different CoT length (1k–4k tokens). We define the **Attention Ratio** as the ratio between (i) the sum of attention weights on harmful tokens *in the prompt* and (ii) those on puzzle tokens *in the prompt*:

$$\text{AttnRatio} = \frac{\sum\limits_{t \in H} \alpha_t}{\sum\limits_{t \in P} \alpha_t} \tag{2}$$

where α_t is the attention weight on token t, H denotes the set of harmful instruction tokens, and P denotes the set of benign puzzle tokens. By fixing the prompt segments and their token counts, this metric isolates the effect of CoT length on attention allocation.